

Algebra II with Trigonometry

Chapter 11.3

Study Guide (Gemini)

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[4666324_alg-2trig-h-chp-11-3-note-docx.pdf](#)

Here is a comprehensive overview, study guide, and practice set based on the provided document about hyperbolas.

1. Short Summary

This document introduces **hyperbolas** centered at the origin. It defines a hyperbola geometrically based on the constant difference of distances from a point on the curve to two fixed points (the foci). The text outlines the standard equations, properties (vertices, foci, transverse axis, and asymptotes), and graphs for both horizontally and vertically oriented hyperbolas. It concludes with a step-by-step method for sketching hyperbolas and provides examples for analyzing equations and deriving equations from given conditions.

2. Core Definitions, Theorems, and Formulas

Core Definitions:

- **Hyperbola:** The set of all points in a plane where the *difference* of whose distances from two fixed points (foci) is a constant.
$$|d(P, F_1) - d(P, F_2)| = 2a.$$
- **Center:** The midpoint between the vertices (and foci). In this text, the center is always at the origin $(0, 0)$.
- **Vertices:** The points where the hyperbola intersects its axis of symmetry.
- **Transverse Axis:** The line segment joining the two vertices. Its length is always $2a$.
- **Branches:** The two separate curves that make up the hyperbola.

- **Asymptotes:** The intersecting lines that the branches of the hyperbola approach as they move further from the center.
- **Central Box:** A sketching tool—a rectangle centered at the origin with sides crossing the axes at $\pm a$ and $\pm b$. Its diagonals extend to form the asymptotes.

Core Formulas:

- **Foci Equation (Pythagorean Relationship):** $c^2 = a^2 + b^2$ (used to find the distance c from the center to the foci).

Standard Equations & Properties: | Property | Horizontal Hyperbola | Vertical Hyperbola | | :--- | :--- | :--- | | **Standard Equation** | $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ | $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$ | | **Vertices** | $(\pm a, 0)$ | $(0, \pm a)$ | | **Foci** | $(\pm c, 0)$ | $(0, \pm c)$ | | **Asymptotes** | $y = \pm \frac{b}{a}x$ | $y = \pm \frac{a}{b}x$ | | **Transverse Axis** | Horizontal (on x -axis) | Vertical (on y -axis) |

(Note: In all cases, $a > 0$ and $b > 0$.)

3. Worked Study Guide: Step-by-Step

Concept 1: Determining Orientation The most important first step when looking at a hyperbola's equation is finding its orientation. Look at which term is **positive**.

- If the x^2 fraction is positive (e.g., $\frac{x^2}{9} - \frac{y^2}{4} = 1$), the hyperbola opens horizontally (left and right). The vertices and foci are on the x -axis.
- If the y^2 fraction is positive (e.g., $\frac{y^2}{9} - \frac{x^2}{4} = 1$), the hyperbola opens vertically (up and down). The vertices and foci are on the y -axis.
- *Note:* The value of a^2 is always the denominator of the positive term.

Concept 2: Extracting a , b , and c Once you know the orientation: 1. Find a : Take the square root of the denominator of the positive term. 2. Find b : Take the square root of the denominator of the negative term. 3. Find c : Use the formula $c^2 = a^2 + b^2$.

- *Why we care:* a gives you your vertices, b helps you draw the central box/asymptotes, and c gives you your foci.

Concept 3: Sketching the Hyperbola Let's apply the document's 4-step sketching method to a generic horizontal equation like $\frac{x^2}{9} - \frac{y^2}{4} = 1$ (where $a = 3, b = 2$):

1. **Central Box:** Plot points at $x = \pm 3$ and $y = \pm 2$. Draw a rectangle connecting these boundaries.
2. **Asymptotes:** Draw diagonal lines straight through the corners of this central box, passing through the origin. These are the lines $y = \pm \frac{2}{3}x$.
3. **Vertices:** Because x^2 is positive, plot your vertices on the x-axis at $(3, 0)$ and $(-3, 0)$.
4. **Sketch:** Start at the vertex and draw curves that open outward, getting closer and closer to the asymptote lines but never touching them.

Concept 4: Working Backward (From clues to equation) If a problem gives you "Vertices at $(\pm 2, 0)$ and Foci at $(\pm 6, 0)$ ":

1. Notice the numbers are in the x -coordinate. The hyperbola is horizontal.
2. The standard form will be $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.
3. Vertices give you $a = 2$, so $a^2 = 4$.
4. Foci give you $c = 6$, so $c^2 = 36$.
5. Use $c^2 = a^2 + b^2$ to find b^2 . $\rightarrow 36 = 4 + b^2 \rightarrow b^2 = 32$.
6. Final equation: $\frac{x^2}{4} - \frac{y^2}{32} = 1$.

4. Practice Problems

These problems are pulled directly from the unsolved examples provided on pages 3 and 4 of the document.

Problem 1: Find the vertices, foci, asymptotes, and length of the transverse axis for the hyperbola: $\frac{x^2}{9} - \frac{y^2}{64} = 1$.

Problem 2: Find the vertices, foci, asymptotes, and length of the transverse axis for the hyperbola: $\frac{y^2}{36} - \frac{x^2}{4} = 1$.

Problem 3: Find the standard equation for the hyperbola that satisfies the given conditions: Foci $(\pm 6, 0)$ and Vertices $(\pm 2, 0)$.

Problem 4: Find the standard equation for the hyperbola that satisfies the given conditions: Vertices $(\pm 1, 0)$ and Asymptotes $y = \pm 5x$.

Problem 5: Find the standard equation for the hyperbola that satisfies the given conditions: Foci $(0, \pm 1)$ and length of the transverse axis is 1.

Answers to Practice Problems

Answer 1:

- Orientation: Horizontal (x^2 is positive).
- $a^2 = 9 \rightarrow a = 3$. $b^2 = 64 \rightarrow b = 8$.
- $c^2 = 9 + 64 = 73 \rightarrow c = \sqrt{73}$.
- **Vertices:** $(\pm 3, 0)$
- **Foci:** $(\pm \sqrt{73}, 0)$
- **Asymptotes:** $y = \pm \frac{8}{3}x$ (using $y = \pm \frac{b}{a}x$)
- **Transverse Axis Length:** $2a = 2(3) = 6$

Answer 2:

- Orientation: Vertical (y^2 is positive).
- $a^2 = 36 \rightarrow a = 6$. $b^2 = 4 \rightarrow b = 2$.
- $c^2 = 36 + 4 = 40 \rightarrow c = \sqrt{40} = 2\sqrt{10}$.
- **Vertices:** $(0, \pm 6)$
- **Foci:** $(0, \pm 2\sqrt{10})$
- **Asymptotes:** $y = \pm \frac{6}{2}x \rightarrow y = \pm 3x$ (using $y = \pm \frac{a}{b}x$)
- **Transverse Axis Length:** $2a = 2(6) = 12$

Answer 3:

- Coordinates are on the x-axis, so it is horizontal.
- Vertices $(\pm 2, 0) \rightarrow a = 2 \rightarrow a^2 = 4$.
- Foci $(\pm 6, 0) \rightarrow c = 6 \rightarrow c^2 = 36$.
- Find b^2 : $36 = 4 + b^2 \rightarrow b^2 = 32$.
- **Equation:** $\frac{x^2}{4} - \frac{y^2}{32} = 1$

Answer 4:

- Vertices are on the x-axis, so it is horizontal.
- Vertices $(\pm 1, 0) \rightarrow a = 1 \rightarrow a^2 = 1$.

- For horizontal, asymptotes are $y = \pm \frac{b}{a}x$. We are given $y = \pm 5x$.
- Therefore, $\frac{b}{1} = 5 \rightarrow b = 5 \rightarrow b^2 = 25$.
- **Equation:** $\frac{x^2}{1} - \frac{y^2}{25} = 1$ (or simply $x^2 - \frac{y^2}{25} = 1$)

Answer 5:

- Foci are on the y-axis, so it is vertical.
- Foci $(0, \pm 1) \rightarrow c = 1 \rightarrow c^2 = 1$.
- Transverse axis length is $2a$. So, $2a = 1 \rightarrow a = 0.5 \rightarrow a^2 = 0.25$ (or $\frac{1}{4}$).
- Find b^2 : $1 = 0.25 + b^2 \rightarrow b^2 = 0.75$ (or $\frac{3}{4}$).
- **Equation:** $\frac{y^2}{1/4} - \frac{x^2}{3/4} = 1$ (or mathematically equivalent: $4y^2 - \frac{4x^2}{3} = 1$)

5. Assumptions and Ambiguities Resolved

- **Standard Form Requirement:** On Page 3, examples *iii* ($x^2 - 3y^2 + 12 = 0$) and *iv* ($9x^2 - 16y^2 = 1$) are not in standard form. I assumed the document intended for students to divide and rewrite these equations so they equal 1 (e.g., example *iv* becomes $\frac{x^2}{1/9} - \frac{y^2}{1/16} = 1$). I omitted these from the basic practice section to focus heavily on the core standard definitions provided on page 2.
- **Image Legibility (Example 2a):** On Page 4, the graph for Example 2a passes through a labelled point $(3, -5)$, but the vertices aren't explicitly labelled with numbers. Based on the tick marks, I assumed the vertex is at $(0, \pm 4)$. Because of this visual ambiguity, I relied on text-based examples (Example 3) to generate the practice problems for clarity.
- **Example 3c Ambiguity:** The text asks for the equation of a hyperbola with asymptotes $y = \pm x$ that passes through $(5, 3)$. Because the coordinates of the point feature a larger x -value than y -value ($5 > 3$), the point lies "outside" the central diagonal cross in the region of the x -axis. This implies a horizontal orientation. I assumed the student would have to deduce this orientation geometrically, but excluded it from the 5 practice problems to avoid testing advanced logic not explicitly explained in the reading.